

CODE SPORT



In this month's column, we discuss a motley collection of interview questions in computer science.

We are at the end of 2018, and as is our regular practice, we close the year with a set of interview questions featuring many of the topics we discussed over the last year. Before we get started on the questions, I wanted to recommend to our readers a couple of books that I think will be useful. A number of readers had asked me for pointers to resources that can provide the requisite mathematics background for machine learning. One of the useful resources is the paper 'The Matrix Calculus You Need For Deep Learning' by Jeremy Howard and Terence Parr, available at <https://explained.ai/matrix-calculus/index.html>. Another is the draft book 'Mathematics for Machine Learning' available at <https://mml-book.github.io/>. Coursera is also offering a specialisation on mathematics for machine learning (<https://www.coursera.org/specializations/mathematics-machine-learning>).

A few readers had also asked me about getting started with deep reinforcement learning. While we have not covered reinforcement learning in our column so far, this is one of the important areas of machine learning. In fact, plenty of machine learning researchers opine that reinforcement learning may very well hold the key to artificial general intelligence (AGI). The best way to get started with reinforcement learning is by following David Silver's lecture series on YouTube and the course slides available at <http://www0.cs.ucl.ac.uk/staff/d.silver/web/Teaching.html>. This can be followed by John Schulman's lectures on deep reinforcement learning which are available on YouTube. The introductory textbook for reinforcement learning is by Sutton and Barto, and is available at <http://incompleteideas.net/book/the-book.html>. We will cover reinforcement learning in the first couple of columns in 2019, in response to many readers' requests. Now let us get started on our list of interview questions.

1. We are all familiar with supervised machine learning (ML) methods, which make the assumption that the training input data satisfies the IID (independent and identically distributed) assumption. Why is this assumption important? Does this assumption hold good for time series data?
2. We are all familiar with vector space word models such as word2vec or Glove. You are given word2vec models for two large corpora, one in English and another in German. Can you come up with an algorithm for machine translation from English to German using these two corpora and word embeddings?
3. Interpretability is a major concern as machine learning models are getting deployed in real-life scenarios such as clinical decision making. There are many definitions of interpretability. However, for our purpose, let us just say that a model's prediction is interpretable if it can be explained in terms of the features of the input data. In classical machine learning models, can you rank the following classification models in the order of increasing interpretability?
(a) Logistic regression, (b) Decision trees, and (c) Random forests.
4. Given that deep learning does not require feature engineering, and learns the features implicitly from input data, how can we make deep learning based classification systems interpretable? Can you explain some methods of interpretability for deep learning classification systems?
5. You are worried that you may be suffering from dengue fever. You know that if someone gets dengue fever, there is a 90 per cent chance that they get a rash on their skin. You also know that dengue is a relatively rare disease, and the possibility of a human getting it is 1 in 100,000. You are